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SNOWMOBILE

Abstract

Embodiments of the present disclosure describe a snowmobile including a chassis, a tunnel, an engine, a drive track and a drivetrain. The tunnel is attached to the chassis and includes running boards located on either side of the tunnel. The engine is attached to the chassis and within an engine bay. The drivetrain is operatively connected to the engine to deliver propulsive power to the drive track, wherein the running boards extend forward alongside at least a portion of the engine bay.

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Background/Summary

[0001] This application is a continuation of U.S. application Ser. No. 18/227,809, filed on Jul. 28, 2023, which is a continuation of U.S. application Ser. No. 17/550,132, filed on Dec. 14, 2021, which is a continuation of U.S. application Ser. No. 16/864,452, filed on May 1, 2020, which is a continuation of U.S. application Ser. No. 15/857,823, filed on Dec. 29, 2017, now U.S. Pat. No. 10,676,157 issued on Jun. 9, 2020, which claims the benefit of U.S. Provisional Application No. 62/441,812, filed on Jan. 3, 2017, and which applications are incorporated herein by reference. A claim of priority to all, to the extent appropriate, is made.

BACKGROUND

[0002] Snowmobiles are popular land vehicles used as transportation vehicles or as recreational vehicles in cold and snowy conditions. Generally, snowmobiles are available for various applications such as deep snow, high performance, luxury touring, and trail riding, for example. In general, a snowmobile has a chassis on or around which the various components of the snowmobile are assembled. Typical snowmobiles include one or more skis for steering, a seat, handlebars, and an endless track for propulsion mounted to a central chassis. The engine drives a ground-engaging endless track disposed in a longitudinally extending drive tunnel. One or more skis serve to facilitate steering as well as to provide flotation of the front of the snowmobile over the snow in which it is operated. A handlebar assembly, positioned forward of the seat, is operatively linked to the skis for steering the snowmobile. The skis may be pivoted to steer the snowmobile, for example, by turning the handlebars.

SUMMARY

[0003] Embodiments of the present disclosure describe a snowmobile that includes a chassis, a tunnel, an engine, a drive track, and a drive train. The tunnel is attached to the chassis and includes running boards located on either side of the tunnel. The engine is attached to the chassis and within an engine bay. The drivetrain operatively interconnects the engine with the drive track for delivering propulsive power to the drive track, wherein the running boards extend forward alongside at least a portion of the engine bay.

[0004] Embodiments also describe a snowmobile that includes a chassis, a tunnel, a seat, an engine, a drive track, a drivetrain, and an exhaust system. The tunnel is attached to the chassis and the seat is attached to the top panel of the tunnel. The engine is attached to the chassis. A drivetrain operatively interconnects the engine with the drive track for delivering propulsive power to the drive track. The exhaust system comprises a first portion extending from the engine, a second portion extending toward the back of the snowmobile in a region between the seat and the top panel of the tunnel, and a third portion extending downward through the tunnel and opening to a region between the tunnel and a top of the drive track.

[0005] Embodiments also describe a snowmobile that includes a chassis, a hood, a bumper, an engine, a drive track, and a drivetrain. The bumper is affixed to the hood to form an integrated hood/bumper assembly. The engine is attached to the chassis. The drivetrain operatively interconnects the engine with the drive track for delivering propulsive power to the drive track.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] This written disclosure describes illustrative embodiments that are non-limiting and non-exhaustive. Reference is made to illustrative embodiments that are depicted in the figures, in which:

[0007] FIG. **1** illustrates a side view **100** of a snowmobile, according to some embodiments.

[0008] FIG. **2** illustrates a side view **200** of a snowmobile with an integrated hood/bumper in an open position, according to some embodiments.

[0009] FIG. **3** illustrates a side view **300** of a snowmobile with the seat removed to show the exhaust system, according to some embodiments.

[0010] FIG. **4** illustrates a top view **400** of a snowmobile, according to some embodiments.

[0011] FIG. **5** illustrates a back view **500** of the front of a snowmobile, according to some embodiments.

[0012] FIG. **6** illustrates a side perspective view **600** of an engine assembly and running boards of a snowmobile, according to some embodiments.

[0013] FIG. **7** illustrates a side perspective view **700** of an engine assembly of a snowmobile, according to some embodiments.

[0014] FIG. **8** illustrates a left side perspective view **800** of a drive line assembly, according to some embodiments.

[0015] FIG. **9** illustrates a perspective view **900** of the snowmobile engine, according to some embodiments.

[0016] FIG. **10** illustrates a perspective view **1000** of a snowmobile chassis, according to some embodiments.

[0017] FIG. **11** illustrates a perspective view **1100** of an air intake filter, according to some embodiments.

[0018] FIG. **12** illustrates a front view **1200** of a snowmobile with the integrated hood/bumper opened to illustrate a cooling system, according to some embodiments.

[0019] FIG. **13** illustrates a perspective view **1300** of the snowmobile with the integrated hood/bumper removed to illustrate the engine and drive line system, according to some embodiments.

[0020] FIG. **14** illustrates a perspective view **1400** of the snowmobile that includes a rack positioned along the integrated hood/bumper, according to some embodiments.

DETAILED DESCRIPTION

[0021] Embodiments of the present disclosure describe a snowmobile in which the engine and drivetrain mounted within an engine bay are defined by a width that is less than a width of running boards located alongside the tunnel of the snowmobile. Configuring the engine and drivetrain within a width less than the width of the running boards allows the running boards to extend, at least partially, alongside the engine bay, extending the forward position of the running boards as compared with typical snowmobile designs. The extended forward position of the running boards allows a rider's feet to be placed further forward along the length of the running boards, which is advantageous in certain riding conditions. In one embodiment, the forward position of the running board is positioned forward of the track drive axis. In one embodiment, the forward position of the running board is located forward of or approximately equal to the center-of-gravity (COG) of the snowmobile. As a result, the COG of the rider is approximately co-planar with the COG of the snowmobile. In some contexts, snowmobile maneuverability is improved by aligning the COG of the rider and the snowmobile.

[0022] Embodiments of this disclosure also describe an exhaust system that extends from the engine to the back of the snowmobile via a path that extends between the seat and the top of the tunnel. Benefits of this exhaust system include reduced width of the exhaust system, which aids in maintaining the width of the engine bay less than the width of the running boards, as well as a

reduction in exhaust noise due to the outlet of the exhaust system being directed downward through the tunnel towards the track.

[0023] Embodiments of this disclosure also describe an integrated hood/bumper component. In one embodiment, the bumper is a structural component that is mounted to the hood and pivotally mounted to the chassis. This configuration allows the integrated hood/bumper to be lifted away from the engine about the pivot point, wherein the bumper supports the hood in the open position, thereby providing easy access to the engine and other components located in the engine bay.

[0024] Referring now to FIGS. 1 and 2, a side view **100** of a snowmobile is shown that includes an integrated hood/bumper in both a closed position and an open position. In the embodiments shown, the snowmobile includes handlebars **102**, steering shaft **103** (see FIG. 2, for example), a seat **104**, skis **106**, a drive track **108**, a chassis/frame **110**, an integrated hood/bumper **111** that includes a hood portion **112** and a bumper portion **114**, an engine **118**, a drive train **120**, and tunnel **122**, which includes running boards **124** extending along the sides of tunnel **122**. The chassis/frame **110** supports the tunnel **122**, the engine **118**, the drive train **120**, the drive track **108**, and skis **106**. The position of the running boards is discussed throughout, as compared with the location of other components of the snowmobile. For purposes of this discussion, the running boards are described with respect to how far forward the running boards **124** extend, with the direction forward referring to the front of the snowmobile. For reference, the forward direction is indicated in a number of the Figures.

[0025] In particular, the embodiment shown in FIG. 1 illustrates the integration of hood portion **112** and bumper portion **114**, and the hinged connection **116** of the integrated hood/bumper **111** to chassis/frame **110**. The hood portion **112** covers and protects various components of the snowmobile. Bumper portion **114** is connected to chassis/frame **110** via hinged connection **116**, and because the bumper portion **114** is structurally connected to the chassis/frame **110**, the bumper portion **114** may be utilized to support the weight of the snowmobile. For example, the bumper portion can be utilized to lift and/or pull the snowmobile.

[0026] The bumper portion **114** can be a single vertical component or a plurality of vertically aligned components (e.g., FIG. 14). The bumper portion **114** components may intersect outside the hood portion **112**, inside the hood portion **112**, or some combination thereof. The bumper portion **114** can be horizontally aligned and may be either a single component or a plurality of components, for example tubular steel or aluminum, though other materials and combinations of materials (e.g., steel or aluminum tubing in combination with a polymeric portion) can also be. As above, the horizontal components may intersect or be integrated at one or more contact points outside the hood portion **112**, inside the hood portion **112**, or some combination thereof. Alternatively, the bumper portion **114** may include individual components that are aligned neither substantially horizontally or vertically, such as in an “X” design. The bumper portion **114** can be rounded, squared, or some other configuration to provide protection from impact, provide a towable connection, or other feature, for example.

[0027] Referring now to FIG. 2, the integrated hood/bumper **111** is shown in the open position, pivoted about hinged connection **116** to expose engine **118** other components located in the engine bay. In the open position, bumper portion **114** can rest on the ground and supports the weight of integrated hood/bumper **111** or, the hood/bumper **111** can be supported by a tether which is attached to a suitable portion of the chassis. In some embodiments, the tether comprises a cable, such as a braided stainless steel cable. The integrated hood/bumper **111** can pivot from about a 45 degree angle (relative to the ground plane) to about a 180 degree angle to provide greater accessibility to the engine bay and related components.

[0028] As discussed in more detail with respect to FIGS. 4-10, the packaging and width of the engine **118** and drivetrain **120** allows the running boards **124** to extend further forward (as indicated by line **126** shown in FIGS. 1-6), relative to the location of the engine **118** and/or the drivetrain **120**. In the embodiment shown in FIGS. 1 and 2, the running boards **124** extend forward

to a position **126** at least partially adjacent to drivetrain **120**, and forward of at least some elements of the drivetrain **120**. As discussed in other embodiments, the forward position of the running boards **124** may also be described with respect to the axis of the track drive (shown in FIG. **6**), the location of the lower A-arms connecting the chassis/frame **110** to the skis **106**, as well as to the COG of the snowmobile. In general, however, a benefit of providing running boards that extend further forward than traditional running boards is it allows a rider's feet to be positioned further forward than in previous designs, which may provide additional rider control in certain riding conditions, for example by centralizing the weight of the rider near the COG of the snowmobile. [0029] Referring now to FIG. **3**, a side view **300** of a snowmobile is shown with the seat **104** and integrated hood/bumper **111** removed to illustrate the location of the exhaust system **302**, according to some embodiments. In the embodiment shown in FIG. **3**, exhaust system **302** includes a first portion **304** that extends from engine **118** to the area between the seat **104** and the top panel of the tunnel **122**, a second portion **306** that extends along the top panel of the tunnel **122** and may include a muffler and/or catalytic converter, and a third portion **308** that extends downward through the tunnel **122** toward the drive track **108**. The seat **104** (shown in FIGS. **1** and **2**), is positioned over the top of the second portion **306**. In some embodiments, the muffler is located under the seat **104**.

[0030] More particularly, in the embodiment shown in FIG. **3**, the first portion **304** of the exhaust system **302** extends from the engine **118** in an approximately forward direction. The first portion **304** curves upward, and around a top side of the drivetrain **120**. As a result, the first portion **304** of the exhaust system **302** does not interfere with the position of the drivetrain **120**, and thus does not contribute to the overall width of the engine **118** and/or drivetrain **120**. Second portion extends along the top of the tunnel **122**, and third portion **308** extends through tunnel **122** and opens towards drive track **108**. In this way, exhaust from the engine is directed along the path defined by exhaust system **302** and is directed or output in the area between the tunnel **122** and the drive track **108**. A benefit of outputting the exhaust in this manner is that the overall noise level of the snowmobile is reduced. Additionally, and as mentioned elsewhere, by moving the exhaust system **302** to a central or interior aligned position, the overall width of the engine bay can be narrowed. [0031] In some embodiments, the engine can be oriented such that the exhaust is directed rearwardly without the first portion **304** that extends in an approximately forward direction. In this way, a reduction in the total length of exhaust pipe can be achieved, resulting in a lighter exhaust system

[0032] Referring now to FIG. **4**, a top view **400** of a snowmobile is shown with integrated hood/bumper **111** and exhaust system **302** removed to illustrate the width of the engine bay **402** relative to running boards **124**. The approximate width of the engine bay is represented as Web , while approximate width of the running boards is illustrated as $w_{sub.rb}$ (outside width of the running boards). Engine bay **402** includes a number of components, including the engine **118**, the drivetrain **120**, and an air filter **404**. In the embodiment shown in FIG. **4**, the engine **118** is a single-cylinder engine, however in other embodiments, engine **118** may be implemented as a two-cylinder engine, four cylinder engine, etc., and may be either a two-stroke or four-stroke type engine. In some embodiments, the center of mass of the engine **118** is offset from a longitudinal centerline plane of the snowmobile, which allows the drivetrain **120** to be positioned so as to reduce the overall width Web of the engine bay. In particular, as illustrated in FIG. **4**, the width Web of the engine bay **402** is less than the width $w_{sub.rb}$ of the running boards **124**. As described in more detail with respect to FIG. **5**, a portion of the running boards **124** may be fitted around one or more engine or driveline components, thereby allowing the running boards **124** to extend forward alongside the engine components, to a forward position indicated by dashed line **126**. In addition, as discussed with respect to the side view shown in FIG. **3**, in one embodiment the routing of the exhaust along the top of the tunnel **122** further reduces the width Web of the engine bay **402**, ensuring this width is less than the width $w_{sub.rb}$ of the running boards **124**.

[0033] The furthestmost position of the running boards **124** as indicated by dashed line **126** determines the furthestmost foot position of the rider. With respect to FIGS. **4-10**, the forward position **126** of the running boards **124** are described with respect to the relative locations of other components. For example, in the embodiment shown in FIG. **4**, running boards **124** extend forward alongside the tunnel **122** to a position approximately equal in the forward direction to the attachment of the lower A-arm **406** to the chassis/frame **110**. In another embodiment, the forward position **126** of the running boards **124** may be described relative to the location of the engine **118** and the drivetrain **120**, wherein the running boards **124** extend forward alongside at least a portion of the engine **118** and the drivetrain **120**, and is forward of the track drive axis illustrated in FIG. **6**. The location of the running boards **124** relative to the engine **118** and drivetrain **120** is in contrast with typical snowmobile designs, in which the drivetrain components and/or exhaust components extend radially outward from that shown in FIG. **4**. As a result, in typical snowmobiles, at least a portion of the drive train components and exhaust components extend to a width that prohibits the running boards from extending as far forward as that illustrated in FIG. **4**.

[0034] In another embodiment, the position of the running boards **124** are described with respect to the center-of-gravity of the snowmobile, indicated by the dot labeled “**410**”. While the center-of-gravity depends on a number of factors, including engine selection, gear selection (e.g., reverse gear required), in the embodiment shown in FIG. **4** the center-of-gravity **410** is located rearward of the forward position **126** of the running boards **124**. As a result, this allows a rider's feet to be positioned forward of or approximately equal to the center-of-gravity **410** of the snowmobile.

[0035] Referring now to FIG. **5**, a back view **500** of the front of a snowmobile is shown with integrated hood/bumper **111** and exhaust system **302** removed to illustrate the relative locations of the engine **118** and drivetrain **120** to the running boards **124**. The view shown in FIG. **5** differs slightly from that shown in FIG. **4**. In particular, while FIG. **4** showed a top view of the snowmobile, the view in FIG. **5** is looking forward from the back of the snowmobile at components included in the front of the snowmobile. For context, steering shaft **103** is illustrated, along with skis **106** and air filter **404**. FIG. **5** illustrates the relative locations of drive train elements, including drive (or primary) clutch **502**, endless, flexible drive belt **503**, driven (or secondary) clutch **504**, jack shaft **506**, and transmission elements—which can be partially or fully enclosed in a case—that include top drive sprocket **508**, cogged belt/chain **510**, and bottom drive sprocket **512**. In some embodiments, the cogged belt or chain **510** can be a gear drive.

[0036] In the embodiment shown in FIG. **5**, engine **118** is a single-cylinder engine, and is offset from the centerline of the snowmobile (e.g., in this embodiment, positioned on the right side of the engine bay). As discussed above, in other embodiments, engine **118** may be implemented with multi-cylinder engines (e.g., two-cylinder, four cylinder). Mechanical power developed by engine **118** is communicated to primary clutch **502** by a drive shaft (not visible). Primary clutch **502** and secondary clutch **504** form a continuously variable transmission (CVT), which acts to communicate the power received from engine **118** to jack shaft **506**. Primary clutch **502** and secondary clutch **504** each include both a stationary sheave and a movable sheave. As the speed of the engine increases, the movable sheaves are actuated toward or away from the stationary sheave, to selectively alter the gearing ratio of the CVT. Power communicated from the CVT to transmission case, which in turn communicates power to the track drive shaft (**802**, FIG. **8**) to provide power to the track. In the embodiment shown in FIG. **5**, the transmission case includes top drive sprocket **508** connected to receive power from the jack shaft **506**, and in turn coupled to bottom drive sprocket **512** by cogged belt/chain **510**. As noted previously, a gear drive can also be used in place of belt/chain **510**. Bottom drive sprocket **512** drives the track drive shaft connected to transfer power to the drive track **108** (shown in FIG. **1**).

[0037] In the embodiment shown in FIG. **5**, the running board **124** includes a gap **520** that allows the bottom sprocket **512** to be located adjacent the running board **124**. The running board continues forward of the bottom sprocket **512**, allowing a rider's feet to be positioned forward of the bottom

sprocket **512** and forward of the drive track shaft (not shown in the view) affixed to the bottom sprocket **512**. In other embodiments, depending on the precise width of the engine **118** and drivetrain **120**, and on the inside width or distance between running boards, the gap **520** may not be necessary, or may be smaller in size. However, as illustrated, the gap **520** does not prohibit a rider from positioning the rider's feet forward of the bottom sprocket **512**.

[0038] Referring now to FIG. **6**, a side perspective view **600** of an engine assembly and running boards of the snowmobile are shown. Visible in this view are the primary clutch assembly **502**, and a secondary clutch assembly **504**. Also, partially visible is mounting bracket **608** (see also FIG. **8**) associated with a distal end of the track drive shaft (not shown) coupled to propel the track drive **108**. Mounting bracket **608** acts to secure the distal end of the track drive shaft to the tunnel **122** of the snowmobile. The axis of the drive track shaft is illustrated by dashed line **604** extending through mounting bracket **608**, while the front of running boards **124** (i.e., front-most location) is illustrated by dashed line **126**. The embodiment shown in FIG. **6** illustrates that the running boards **124** extend forward of the track drive shaft, illustrated by dashed line **604**. As a result, this allows a rider's feet to be positioned forward of the track drive shaft.

[0039] In addition, the forward position of the running boards may also be defined with respect to other structural elements, such as the lower A-arms **406** connecting the skis **106** to the chassis/frame **110**. In the embodiment shown in FIG. **6**, the front of the running boards **124** end at approximately the same location as one leg of the lower—**406** is connected to chassis/frame **110**. In some embodiments, a portion of the running boards **124** can extend forward of a portion of the A-arm.

[0040] Referring now to FIG. **7**, a side perspective view **700** of an engine assembly of a snowmobile is shown. The view shown in FIG. **7** is taken from the side of the snowmobile opposite that shown in FIG. **6**. Visible in this view is the right-side running board **124**, steering shaft **103**, engine **118**, air filter **404**, top sprocket **508**, cogged belt/chain **510**, bottom sprocket **512**, and intake manifold **704**. Once again, the forward position of the running boards **124** is indicated by dashed line **126**, and is located forward of the axis **604** defining the location of track drive shaft (which is driven by bottom sprocket **512**). In addition, FIG. **7** illustrates the use of a small gap **520** to make room in the running board **124** for bottom sprocket **512**, thereby allowing running board **124** to extend forward of bottom sprocket **512**.

[0041] In addition, the embodiment shown in FIG. **7** illustrates the location of brake **516** and brake disc **518** on the drive shaft **506**, located adjacent to sprocket **508**. A benefit of locating brake **516** and brake disc **518** as shown in FIG. **7** is that the brake disc **518** and assembly is easily accessible for maintenance and repair. Moreover, a bracket **702** can locate both the jack shaft and the brake assembly (e.g., caliper).

[0042] Referring now to FIG. **8**, a left side perspective view **800** of the drivetrain **120** is shown, according to some embodiments. Visible in this view is the drive or primary clutch **502**, endless drive belt **503**, driven or secondary clutch **504**, jack shaft **506**, brake assembly **516**, brake disc **518**, cogged belt/chain **510**, lower sprocket **512**, track drive shaft **802**, and track drive sprocket **804**. As discussed above with respect to FIG. **5**, primary clutch **502** and secondary clutch **504** provide a continuously variable transmission (CVT) that transfers mechanical energy developed by in the engine to the transmission case that includes components such as the upper sprocket and lower sprocket **512**, which in turn transfers the mechanical energy to track drive shaft **802**. Track drive sprocket **804** transfers mechanical energy from track drive shaft **802** to the track drive (not shown). Dashed line **604** (shown in FIGS. **6** and **7**) is drawn along the centerline axis of track drive shaft **802**. Additional benefits of this configuration include the length of jack shaft **506** and track drive shaft **802** are reduced, thereby reducing the overall weight and cost of the drivetrain **120**.

[0043] Referring now to FIG. **9**, a perspective view **900** of the snowmobile single-cylinder engine **118** is illustrated, according to some embodiments. The single-cylinder engine **118** shown in FIG. **9** includes a cylinder head assembly **902**, crankcase assembly **904**, air intake port **908**, which is

coupled to intake manifold **704** (see FIG. 7), exhaust port **910**, which is coupled to the exhaust system, and front mounting bracket **906**. A primary drive clutch **502** is also illustrated, connected directly to the drive shaft of the engine **118**. In this embodiment, the position of exhaust manifold **910** allows exhaust system **304** (shown in FIG. 3) to curl around the top of the engine **118** and under the seat **104** of the snowmobile. In some embodiments, the front mounting bracket **906** is a double puck style mounting bracket.

[0044] In other embodiments, single-cylinder engine **118** is replaced with a multi-cylinder engine (e.g., two-cylinder, four cylinder, etc.). In addition, the multi-cylinder engine may be a two-stroke type engine, or a four-stroke type engine. Depending on the dimensions of the engine **118**, the amount of offset from the centerline of the snowmobile may be increased or decreased in order to fit engine **118** and drivetrain **120** within a desired width.

[0045] Referring now to FIG. 10, a perspective view **1000** of the snowmobile chassis/frame **110** is shown, according to some embodiments. Chassis/frame **110** includes the engine bay **402**, located at the front of the snowmobile as well as tunnel **122**. As discussed above, the tunnel includes running boards **124** located on each side of the tunnel **122**. In the embodiment shown in FIG. 10, running boards **124** extend forward to a position at which engine frame support post **1004** and **1006** extend from tunnel **122** toward the engine bay **402**.

[0046] Referring now to FIG. 11, a perspective view **1100** of a cyclonic air intake filter **404** is shown, according to some embodiments. Cyclonic air intake filter includes throttle body **1102**, air inlet **1104**, and filter chamber **1108**. The cyclonic air filter communicates air provided at air inlet **1104** to filter chamber **1108**. Once in the filter chamber **1108**, incoming air within the filter chamber swirls (cyclonic swirling). The swirling action causes heavier particles (e.g., dust, dirt) to be pushed to the outside of the air chamber **1108**, to be collected or discharged. The remaining air swirls through a filter (not shown) located within the filter chamber **1108** and the filtered air is output through throttle body **1102**.

[0047] Referring now to FIG. 12, a front view **1200** of a snowmobile is shown with the hood/shroud opened to illustrate a radiator based cooling system **1202**, according to some embodiments. In particular, in the embodiment shown in FIG. 12, a radiator **1202** is mounted on the front of the engine bay. Heat from the engine is removed via coolant pumped through the engine via a water pump (not shown), which can be engine driven or electric, and into the radiator **1202** to dissipate the heat via the collection of tubes and fins associated with radiator **1202**. In the embodiment shown in FIG. 12, a fan mounted on the radiator blows air over the plurality of tubes/fins to aid in the dissipation of heat. A benefit of utilizing a radiator-based cooling system such as that shown in FIG. 12 as opposed to systems that rely on heat exchangers is the lower cost associated with this type of system as opposed to a system that utilizes heat exchangers. In some embodiments, however, a typical heat exchanger type system may be employed, for example under the tunnel, to dissipate heat generated by the engine **118**.

[0048] Referring now to FIG. 13, a perspective view **1300** of the snowmobile is shown with the hood/shroud removed to illustrate the engine and drive line assembly components included in the engine bay. In particular, the embodiment shown in FIG. 13 illustrates a number of the features discussed individually with respect to FIGS. 3-12. Visible in the embodiment shown in FIG. 13 includes handlebars **102**, steering shaft **103**, skis **106**, chassis/frame **110**, tunnel **122**, running boards **124**, exhaust system **302** (extending away from the engine, but not shown extending along the top of tunnel **122**), cyclonic air filter **404**, primary clutch drive **502**, secondary driven clutch drive **504**, and radiator **1202**. Attached to primary clutch drive **502**, but obscured in this view, is single-cylinder engine **118**.

[0049] Referring now to FIG. 14, a perspective view **1400** illustrates the inclusion of a rack assembly **1402** physically mounted to bumper **114**. As discussed with respect to FIGS. 1 and 2, the embodiment shown in FIG. 14 utilizes an integrated hood/bumper **111**. In the embodiment shown in FIG. 14, a rack **1402** is mounted to the bumper assembly **114**. In particular, rack **1402** provides a

frame to which objects or gear may be attached to and retained by. Rack **1402** may be removably attached to bumper **114**, or may be integrated as part of bumper **114**.

[0050] While the disclosed snowmobile has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A snow vehicle, comprising: a tunnel including a first side panel, a second side panel, and a rear end; a first support member positioned outboard the first side panel; a second support member positioned outboard the second side panel; and a bumper positioned at the rear end of the tunnel, wherein the first support member and the second support member are secured to the bumper, wherein the first support member, the second support member, and the bumper define a continuous rail along the first side panel, the second side panel, and the rear end of the tunnel.
2. The snow vehicle of claim 1, wherein the first side panel and the second side panel include a bottom edge, wherein the first support member, the second support member, and the bumper extend along the bottom edge.
3. The snow vehicle of claim 1, wherein the first side panel and the second side panel vertically extend between a top panel and the continuous rail, wherein no portion of the first side panel or the second side panel extends below a bottom surface of the continuous rail.
4. The snow vehicle of claim 1, wherein the first support member, the second support member, and the bumper define a tubular shape, wherein the continuous rail defines a continuous tube along the first side panel, the second side panel, and the rear end of the tunnel.
5. The snow vehicle of claim 1, wherein the bumper includes a first bumper arm positioned on the first side panel and a second bumper arm positioned on the second side panel.
6. The snow vehicle of claim 5, wherein the first support member and the first bumper arm form a first continuous rail along an entire length of the first side panel and wherein the second support member and the second bumper arm form a second continuous rail along an entire length of the second side panel.
7. The snow vehicle of claim 1, wherein the first support member is spaced apart from the first side panel and wherein the second support member is spaced apart from the second side panel.
8. The snow vehicle of claim 1, wherein the first support member includes a first running board and wherein the second support member includes a second running board.
9. The snow vehicle of claim 8, wherein the first running board and the second running board are secured to a bottom surface of the continuous rail.
10. The snow vehicle of claim 1, wherein the first support member and the second support member are secured to a rear suspension component of the snow vehicle.
11. A snowmobile, comprising: a chassis; a tunnel including a first side panel, a second side panel, and a rear end; a first running board positioned outboard the first side panel; a second running board positioned outboard the second side panel; an engine coupled to the chassis, the engine including a crankcase including a crankshaft axis of rotation; a drive track; and a drivetrain operatively connecting the engine with the drive track for delivering propulsive power to the drive track, wherein the first running board and the second running board extend forward alongside the crankshaft axis of rotation.
12. The snowmobile of claim 11 further comprising a first ski coupled to the chassis and a second ski coupled to the chassis.

13. The snowmobile of claim 12, wherein the first running board and the second running board extend forward of at least a portion of the first ski and the second ski.

14. The snowmobile of claim 12, wherein the first ski is coupled to the chassis via a first A-arm and wherein the second ski is coupled to the chassis via a second A-arm, wherein the first running board extends forward of a first connection point between the first A-arm and the chassis and wherein the second running board extends forward of a second connection point between the second A-arm and the chassis.

15. The snowmobile of claim 12, wherein the first running board extends outboard of at least a portion of the first ski and wherein the second running board extends outboard of at least a portion of the second ski.

16. The snowmobile of claim 11 further comprising a bumper positioned at the rear end of the tunnel, wherein the first running board and the second running board are secured to the bumper, wherein the first running board, the second running board, and the bumper define a continuous rail along the first side panel, the second side panel, and the rear end of the tunnel.

17. A snowmobile, comprising: a chassis; a tunnel including a first side panel and a second side panel; a first running board positioned outboard the first side panel and a second running board positioned outboard the second side panel; a first ski coupled to the chassis and a second ski coupled to the chassis; an engine coupled to the chassis; wherein the first running board and the second running board extend forward of at least a portion of the first ski and the second ski.

18. The snowmobile of claim 17, wherein the first ski is coupled to the chassis via a first A-arm and wherein the second ski is coupled to the chassis via a second A-arm, wherein the first running board extends forward of a first connection point between the first A-arm and the chassis and wherein the second running board extends forward of a second connection point between the second A-arm and the chassis.

19. The snowmobile of claim 17, wherein the first running board extends outboard of at least a portion of the first ski and wherein the second running board extends outboard of at least a portion of the second ski.

20. The snowmobile of claim 17 further comprising: a drive track; and a drivetrain operatively connecting the engine with the drive track for delivering propulsive power to the track drive, wherein the engine includes a crankcase including a crankshaft axis of rotation, wherein the first running board and the second running board extend forward alongside the crankshaft axis of rotation.
